



SET-1

Lab Evaluation-I

Data Analytics Using Python (CSE2101)

Total Marks= 20 [CO2]

Date: Sep 9, 2025

Duration: 60 mins

Name: _____

Roll No: _____

PC No: _____

Section: _____

Instruction: Save your jupyter notebook with your name, last 3 digits of enrollment no and section. Eg: Your name is "Kiran", Enrollment No is "123456" and section is "CSE1" then save your file as "**Kiran_456_CSE1**"

A mid-sized organization wants to create a synthetic dataset to better understand workforce patterns and run internal analytics. To do this, the HR analytics team decides to generate a dataset of employees capturing key details about their demographics, roles, and performance.

Each employee is uniquely identified by an **Employee_id** ranging from 1 to the total number of employees. The **Department** of each employee is randomly assigned from *Sales*, *HR*, *IT*, *Finance*, and *Marketing* with equal probability, reflecting the company's diverse functional areas.

The **Age** of employees is uniformly distributed between 22 and 60 years, representing a mix of fresh graduates and experienced professionals. Each employee also has a **Hire_date**, randomly selected between *2010-01-01* and *2022-12-31*, covering a realistic hiring timeline over the past decade.

Compensation details are captured through **Salary**, which follows a normal distribution with an average of ₹60,000 and a standard deviation of ₹15,000, clipped between ₹30,000 and ₹150,000 to avoid unrealistic values. Performance is measured using a **Performance_score**, assigned as integers between 1 and 10.

Finally, the company records employment type with the **Full_time** column, where 80% of employees are full-time and 20% are part-time.

The resulting dataset, stored in a DataFrame named **df_employee**, provides a realistic synthetic workforce profile for HR managers to analyze salary trends, departmental distribution, employee performance, and workforce composition.

(8)

Perform the following queries:

1. What is the average salary of each department? Display and visualize the results with appropriate plot. **(1+2)**
2. What percentage of full-time and part-time employees works in the company. Display and visualize the results with appropriate plot. **(1+2)**
3. Visualize the distribution of salaries across departments to compare spread and detect potential outliers. **(2+1)**
4. How many employees were hired each year. Display the result. **(2+1)**

Note: Make sure all plots should have proper aesthetics and clearly readable.



SET-2

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A retail company is analyzing its product portfolio across different categories to understand sales, customer feedback, and stock availability. To simulate this study, the company decides to generate synthetic data for its products.

Each product is uniquely identified by a **Product_ID** ranging from 1 to the total number of products. The **Category** of each product is randomly assigned from *Electronics*, *Clothing*, *Groceries*, and *Furniture*, with all categories being equally likely. The **Price** of products follows a normal distribution with an average of ₹500 and a standard deviation of ₹200, but prices are restricted to lie between ₹100 and ₹2000 to reflect realistic market values.

Sales performance is tracked through **Quantity_Sold**, which varies randomly between 10 and 500 units. The **Launch_Date** of each product falls within the period from 2015-01-01 to 2023-12-31, representing the timeline of when products were introduced to the market. Customer satisfaction is represented by **Rating**, taking values between 1 and 5. Finally, the **In_Stock** status indicates current availability, with 70% of products marked as available (True) and 30% as out of stock (False).

The resulting dataset, stored in a DataFrame named **df_product**, provides a comprehensive synthetic view of the company's product line for further analysis. (8)

Perform the following queries:

1. Compute the average price of products per category and display it. Plot the result to compare price distribution across categories. **(1+2)**
2. Display the total quantity sold per year and visualize it. **(1+2)**
3. Display the relationship between product *Price*, *Quantity_Sold*, and *Rating*. Visualize the result with appropriate plot. Interpret the strongest relationship observed. **(1+1+1)**
4. Identify the year with maximum total sales revenue ($Price \times Quantity_Sold$). Display the result. **(2+1)**

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SET-3

Lab Evaluation-I

Data Analytics Using Python (CSE2101)

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Urban Bites is a popular casual-dining restaurant in Gurugram that wants to study customer behavior, order patterns, and revenue trends. To prepare for a data-driven strategy, the management decides to generate synthetic data representing customer groups visiting over a month.

Each group of customers is uniquely identified with a **Group_ID** ranging from 1 to the total number of groups. The **Group_Size** varies randomly between 1 and 6, since tables can accommodate up to six guests.

Guests can order from four main **Food_Categories**: *Starters, Main Course, Desserts, and Beverages*. Each order is randomly assigned but no guest orders more than one main dish. The **Bill_Amount** for each group is calculated based on menu prices, with starters priced between ₹100–₹250, main course items between ₹200–₹600, desserts between ₹80–₹200, and beverages between ₹50–₹180.

Each group is also assigned a **Visit_Date**, randomly chosen within the month of August 2023, to analyze daily footfall. In addition, a **Payment_Mode** is recorded, randomly assigned from *Cash, Card, UPI, or Wallet*, with UPI being the most common.

Customer satisfaction is captured through a **Rating** column (1–5 scale), reflecting post-meal feedback. The restaurant also records whether a group availed any **Promotional_Offer**, with a 30% chance of True (Yes).

The resulting dataset, stored in a DataFrame named **df_restaurant**.

(10)

Perform the following queries along with visualization:

1. Which days of the week bring in the most revenue? Visualize the results. **(1+2)**

2. Identify and display popular food categories across different group sizes? Visualize the results. **(1+2)**

3. Identify and display Customer satisfaction trends with and without offers. Visualize the results. **(1+2)**

4. Find the revenue share by payment mode. Visualize the results. **(1+2)**

Note: Make sure all plots should have proper aesthetics and clearly readable.



SET-4

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Wellness Care Hospital in Rohtak is building a digital record system to better understand **outpatient trends** and to support doctors in decision-making. To achieve this, the hospital generates a dataset called **df_opd** that records important information about each patient visit.

Every **Patient_ID** is unique so that each patient can be tracked properly. The hospital keeps record of the **Age** of patients (ranging from 5 to 90 years), as both children and elderly people frequently visit the OPD. The **Gender** of the patient (Male or Female) is also recorded for demographic and treatment purposes.

Patients are assigned to a **Department** depending on their health concern. The OPD has four main departments – *Cardiology*, *Orthopedics*, *Neurology*, and *General Medicine*. For each patient, the **Visit_Date** is recorded, covering all days from **January 1, 2023, to December 31, 2023**, so that seasonal and monthly patterns can be studied.

Doctors also record each patient's **Systolic_BP** (systolic blood pressure), with values ranging between **90 and 180**, since abnormal readings often reveal early signs of conditions like hypertension. Finally, the hospital keeps track of the **Bill_Amount**, which includes consultation charges, basic medicines, and diagnostic tests. This amount is modeled as normally distributed with an average of ₹2,500 and a standard deviation of ₹800, but always lies between ₹500 and ₹8,000.

The resulting dataset, stored in a DataFrame named **df_hospital**.

(8)

Perform the following queries along with visualization:

1. Which department earns the most revenue? Visualize the results. **(1+2)**
2. What is the distribution of patient age and gender across departments? Visualize the results. **(1+2)**
3. How do blood pressure levels vary across different age groups? Visualize the results. **(1+2)**
4. What are the monthly visit and revenue trends in 2023? Visualize the results. **(1+2)**

Note: Make sure all plots should have proper aesthetics and clearly readable.